Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/03/2021
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Project Title:	Microgravity Studies of Nanoparticles and Topological Defects in Liquid Crystal Thin Films		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/02/2016	**End Date:**	12/01/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:	ISS		
Flight Assignment:	Liquid Crystal Facility NOTE: End date changed to 12/1/2022 per NSSC information (Ed., 1/26/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lacaze, Emmanuelle Ph.D. (Université Pierre et Marie Curie, France) Hegman, Torsten Ph.D. (Kent State University) Taylor, Philip Ph.D. (Case Western Reserve University) Lemieux, Robert Ph.D. (COLLABORATOR / University of Waterloo)		
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Task Description:	Nanoparticle / liquid crystal composites have created new avenues for scientific discovery and technological development. For example, recent studies have shown that mixtures of liquid crystals and appropriate semiconducting nanoparticles can be exploited for potential high efficiency and easily manipulatable photovoltaic devices. But the development of this and other technologies requires a fundamental understanding of how the nanoparticles incorporate and move within the liquid crystal, how they aggregate in and around topological defects, and how the nanoparticles affect the defects themselves. We have shown that fluorescent carbon dots (FCDs) may be dispersed in liquid crystals and used as a probe of defects, demonstrating that they accumulate in regions of high liquid crystal curvature so as to relieve the local strain. As a result their emitted light can serve as a quantitative indicator for the interaction and evolution of topological defects. Interpretation of these initial results, however, is compromised by the presence of substrates, which is required in an Earth-bound environment. Here we will perform experiments in the absence of a container by utilizing the Observation and Analysis of Smectic Islands In Space (OASIS) module aboard the International Space Station (ISS). By studying thin flat liquid crystal films doped with FCDs -- this will be done by dissolving the FCDs in a solvent and "shooting" slowly moving pico droplets of the mixture at the liquid crystal film -- in a microgravity environment, we will obviate the need for bounding substrates, thereby rendering a much simpler system for interpretation. In particular, we will use OASIS' cameras to study the spatial distribution and transport of the FCDs in and around topological defects in the smectic-C phase of the liquid crystal, especially close to the smectic-A phase transition temperature. A related project will be undertaken in which the film will be bombarded by droplets of methanol solvent from a pico inkjet printer involving a large mechanical impulse. The impact of the droplets will induce the formation of "islands" and "divots" of smectic liquid crystal. The goal of this part of the project will be to determine the size and thickness of these induced structures as a function of the film thickness and the droplet speed. The work will involve both experimental and theoretical components. The results will provide a clear picture of how nanoparticles distribute themselves over time, how defects in the liquid crystal interact, and how islands form on a liquid crystal film.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid crystal (LC) textures and topological defects have been of interest for well over a century. Defects in two-dimensions generally are categorized in terms of the "defect strength"-- "s". Studies of defects can be used to obtain values of LC anchoring strength coefficients and elastic constants, albeit with limited accuracy. Defects are central in establishing the director profile in geometrically-confined liquid crystals, and serve as an important signature when determining the symmetry of phases, for example, distinguishing between a biaxial nematic phase and a Smectic-C (Sm-C) phase. Defect dynamics provide another important field of study, as defect motion is extremely sensitive to boundary effects and provides information about surfaces and impurities. Island and divot formation are useful in understanding the layer structure of the smectics and how layers reorganize. They provide information about the surface tension of the liquid crystal. Perhaps most important use of topological defects is that they can serve as traps for micro- and nanoparticles (NPs), as the presence of the particles reduces or eliminates the divergent energy associated with the defect core. In other words, nanoparticles (NPs) migrate within the LC in the direction of increasing stress associated with the liquid crystal's orientational order, as the presence of the particles provides relief from the large stresses that would be present at LC singularities. The incorporation of NPs into liquid crystals and their subsequent behavior in and around defects finds applications in areas such as energy harvesting, photonic structures, mirrorless lasing, micro-patterned LC devices, and chiral metamaterials for photonics and communications applications. They also may have applications in pharmaceuticals, as an alternative means of drug delivery.
Task Progress:	The SARS-CoV-2 virus and resulting Covid-19 had a deleterious effect on our effort. Much of our work has been calculational, along with limited laboratory work. Much of our effort went into working with the groups in Boulder and Kent, in conjunction with the NASA Glenn team, on the Science Requirements Document (SRD). Continued from the previous year, we have been investigating film thickness measurement techniques, and have focused on an approach involving illumination with a quasi-white light source (but having a known spectral distribution) and imaging of the reflected light using the R, G, and B channels of a CCD (charge-coupled device) camera, also of known spectral response. Previously, we modeled the intensity vs. film thickness for each of the channels, and developed a calculational technique to extract the film thickness and avoid the ambiguity of an intensity that is periodic in film thickness by using multiple color sensors in the detector. Moreover, this approach is superior to the laser illumination approach as it allows us to image an area, and extract the local film thickness when islands, divots, or steps in thickness are present. An automated image processing macro was developed to extract intensity values from film images. This work was conducted partially at Case and partially remotely during the spring of 2021. We investigated the use of one or multiple LEDs (light-emitting diodes) for illumination, which would obviate the need to account for the illumination spectrum of the quasi-white light. Reflection intensity measurements are being conducted on films of various thickness that contain steps and islands, and have spent considerable time accurately measuring the refractive indices of the relevant liquid crystals at appropriate temperatures and wavelengths. Using Dr. Tin's film holder, we have been drawing films at various spreader speeds in order to assess empirically how the approximate film thickness and stability is related to the speed. We have found that there is little correlation between spreading speed and film thickness, suggesting that the actual flight experiments will need to be performed by creating a film of arbitrary thickness, measuring the thickness, performing the experiment, breaking the film, and then creating another film of arbitrary thickness. A patterned surface defect of strength m = +1 and its associated disclination lines can decompose into a pair of surface defects and disclination lines of strength m = +½. For a negative dielectric anisotropy liquid crystal subjected to an applied ac electric field E, these half-integer defects are observed to wobble azimuthally for E > some threshold field and, for sufficiently large fields, to co-revolve antipodally around a central point approximately midway between the two defects. This behavior is elucidated experimentally as a function of applied field strength E and frequency, where the threshold field for full co-revolution scales as sqrt (frequency). Concurrently, nematic electrohydrodynamic instabilities were investigated. A complete field vs. frequency "phase diagram" compellingly suggests that the defects' fluctuations and eventual co-revolutions are coupled strongly to the presence of the hydrodynamic instability, resulting in a Lehmann-like mechanism that drives the co-revolution.
Bibliography Type:	Description: (Last Updated: 12/04/2024)

Articles in Peer-reviewed Journals	Susser AL, Kralj S, Rosenblatt C. "Co-revolving topological defects in a nematic liquid crystal." Soft Matter. 2021 Oct 8;17:9616-23. https://doi.org/10.1039/D1SM01124C , Oct-2021